

1. Input Channel vs. Output Channel in a Convolutional Layer:

- A. input channel: the distinct feature maps produced by the preceding convolutional layer that are fed into the current convolutional layer, (or the multiple channels of an image).
- B. output channel: the new feature maps designed and produced by the current convolutional layer itself.

2. Decoupling of Channel Numbers:

- A. the num of input channels in a convolutional layer is completely decoupled from the num of output channels in that same convolutional layer.
- B. the num of input channels affects the num of weights for a single neuron in a convolutional layer.
- C. the num of output channels affects the total num of neurons in a convolutional layer.

3. Stacking of Convolutional Layer:

- A. by stacking multiple convolutional layers from bottom to top, the output channels of the current convolutional layer serve as the input channels of the next convolutional layer. therefore, if a convolutional layer has C output channels, the subsequent layer will have C input channels. that is, a single neuron in the subsequent layer will have $K \times K \times C$ input weights and $K \times K \times C$ data inputs.

B. the bottom convolutional layer:

$\Rightarrow$ num of input channels = 3, it directly receives 3 color channels (Red, Green, Blue) of the image.

$\Rightarrow$ num of output channels = 8, the layer produces total of 8 feature maps.

the top convolutional layer:

$\Rightarrow$ num of input channels = 8, because the previous layer's output channels directly become this layer's input channels, it receives the 8 feature maps.

$\Rightarrow$ num of output channels = 4, this layer combines the lower-level feature maps into 4 higher-level component feature maps

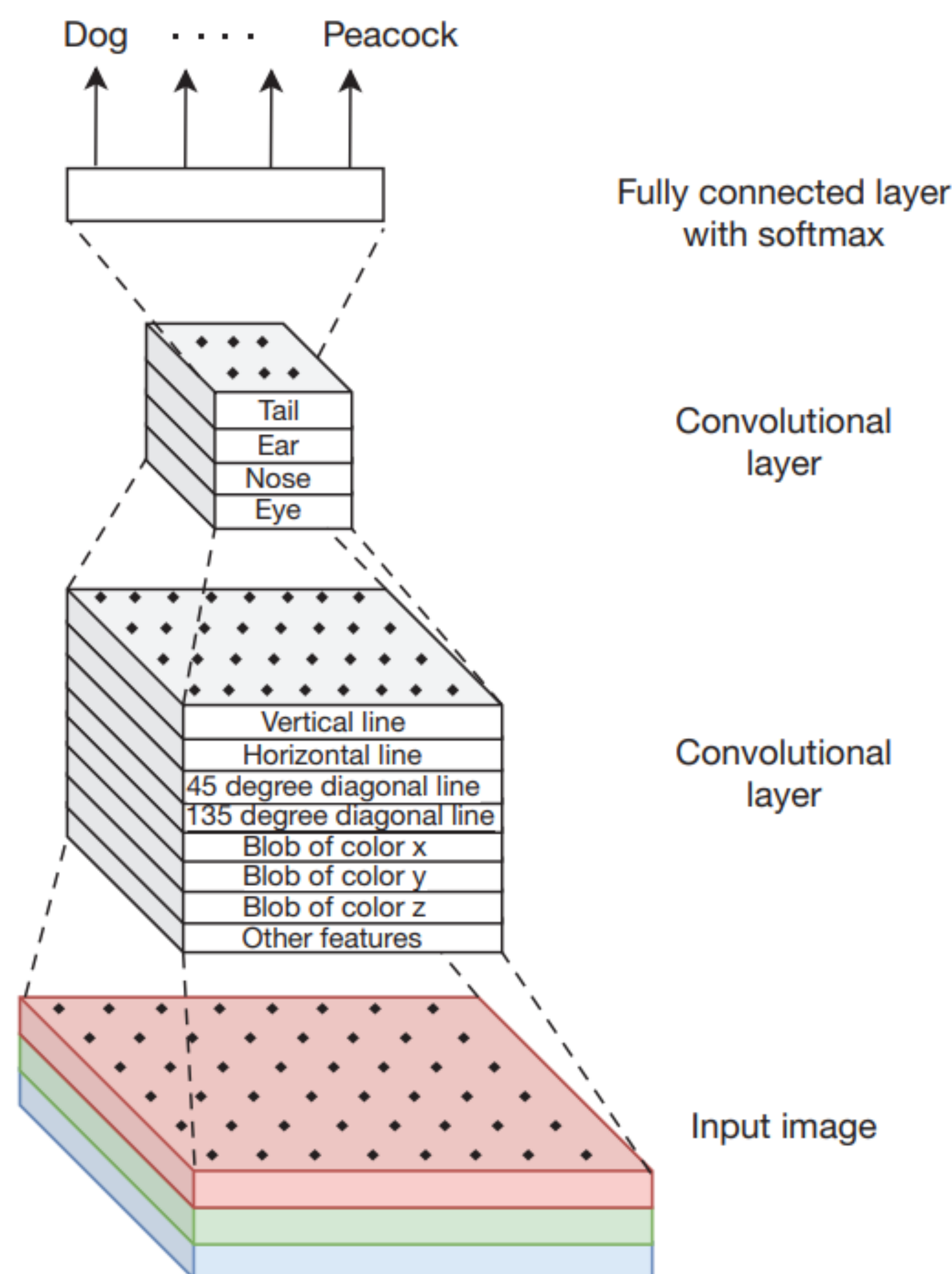


Figure 7-7 Convolutional neural network built from two convolutional layers and one fully connected layer